

Prakyat Prakash

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EDUCATION

Rochester Institute of Technology

Rochester, NY

Masters in Computer Science

Aug 2024 – Dec 2026

- **Relevant Coursework:** Data Structures and Algorithms, Computer Architecture, Machine Learning, Foundations of Artificial Intelligence, Big Data, Advanced Computer Vision

PUBLICATIONS

Pizzi, J.R., Adhikari, I., **Prakash, P.**, Miyamoto, H., & Cui, F. (2026). **Machine learning-based determination of sex-related bladder cancer biomarkers.** *Frontiers in Bioinformatics*, 6. <https://doi.org/10.3389/fbinf.2026.1794098>.
Carson, C.W., Nagalakshmi, S.U., Adhikari, I., Freewoman, J.M., Pizzi, J.R., **Prakash, P.**, & Cui, F. (2025). **Intrinsic DNA codes govern distinct modes of nucleosome-transcription factor interactions.** *bioRxiv*.
<https://doi.org/10.64898/2025.12.30.697010>.

EXPERIENCE

Graduate Research Assistant

May 2025 – Present

Machine Learning Research Assistant

Rochester, NY

- Co-authored 2 research papers in computational genomics, including a peer-reviewed publication in *Frontiers in Bioinformatics* on ML-based bladder cancer biomarker discovery and a paper in *Nature Communications* on nucleosome-transcription factor interactions
- Engineered an end-to-end PTM verified negatives extraction pipeline processing 24.4M raw peptide-spectrum matches into 361,789 verified phosphosites across 35 parallel workers, replacing the field-standard assumption that all non-positive sites are negatives.
- Built a dual-pass FASTA verification pipeline with tryptic context filtering to extract true experimentally confirmed negatives from 5 large biological databases, correcting trypsin-cleavage bias and reducing terminal-K over-representation from 33.2% to 3.6%

Acinonyx Technologies Private Limited

Mar 2024 – Jun 2024

AI Intern

Bengaluru, India

- Built an AI-driven backend service using Flask and REST APIs to automate license management, reducing manual oversight by 30% and streamlining renewals for 500+ users
- Developed a ML model to predict license expiration patterns, enabling real-time tracking and cutting unexpected service downtime by 40%.
- Applied AI-powered analytics on transactional data to trigger automated payment reminders, improving overall processing efficiency and driving measurable sales growth.

PROJECTS

PTM Dataset Extraction Pipeline | *Pandas, NumPy, UniProt* | *Live Dashboard*

Dec 2025 – March 2026

- Reconstructed phosphorylation and lysine acetylation training datasets from 5 mass spectrometry repositories sourced from the DeepMVP/PTMAtlas compendium, processing 24.4M raw peptide-spectrum matches into 361,789 tryptic-verified unique phosphosites across S, T, and Y residues.
- Replaced the field-standard assumption that all non-positive sites are negative by extracting true negatives from experimentally observed unmodified residues, implementing dual-pass FASTA verification with tryptic context filtering across 35 parallel workers.
- Corrected trypsin-cleavage bias in acetylation data via two-stage K-position downsampling, reducing terminal-K over-representation from 33.2% to 3.6%.

NYC Employee Time and Pay Dashboard | *Python, AWS Lambda, DynamoDB, API Gateway, S3*

Ongoing

- Built a serverless payroll system on AWS implementing a NY State labor law rules engine handling 8+ compliance rules including overtime, holiday pay, and spread-of-hours across 500+ shifts
- Designed a REST API with 5 Lambda-backed endpoints delivering real-time pay stub generation in under 2 seconds.
- Integrated Spark for batch processing 10,000+ historical payroll records, cutting weekly reporting time by 60%

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL (Postgres), JavaScript, HTML/CSS

Frameworks: React, Node.js, Flask, FastAPI, Streamlit, TensorFlow, Scikit-learn, Pandas, OpenCV

Machine Learning: Deep Learning, Bayesian Neural Networks, NLP, Computer Vision, Pipeline Development

Developer Tools: Git, VS Code, PyCharm, Jupyter Notebook, IntelliJ, Postman

Cloud and Data: AWS (Lambda, DynamoDB, API Gateway, S3, CloudWatch, IAM, Bedrock), Apache Spark, Docker